

E-Commerce Sales Performance Analysis

Business Report

Analysis Period: January–June 2026

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1. Executive Summary

This report evaluates e-commerce sales performance from January to June 2026 using transaction-level sales data covering **1,000 orders, 3,035 units and approximately ₹27.62 million in revenue**.

The analysis shows that revenue is highly concentrated across a small number of categories and products. **Electronics generated approximately 78.2% of total revenue**, while the top five products contributed approximately **88% of total revenue**.

Geographically, Pune, Mumbai, Jaipur and Bangalore were among the strongest-performing cities, while Hyderabad and Ahmedabad generated substantially lower revenue than the leading cities.

Revenue reached its highest level in January at approximately **₹5.21 million** but declined by approximately **22% from May to June**, with June revenue of approximately **₹3.81 million**.

A short-term revenue forecasting model was also developed. The selected Random Forest model achieved an MAE of approximately **₹92,069** and generated a seven-day expected revenue of approximately **₹1.27 million**.

The findings indicate opportunities to reduce dependence on a small number of products, investigate the June revenue decline, improve performance in underperforming cities, and use forecasting as an additional planning input.

2. Business Objective

The purpose of this analysis is to answer five key business questions:

1. How is the business performing overall?
2. Which products and categories generate the most revenue?
3. Which cities are performing strongly or weakly?

4. Where are the major revenue trends and risks?
 5. What actions can improve future sales performance?
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3. Business Performance Overview

KPI	Performance
Total Orders	1,000
Total Units Sold	3,035
Total Revenue	₹27.62M
Average Order Value	₹27,620.65
Average Selling Price	₹8,940.49

The business generated approximately **₹27.62 million in revenue** during the six-month analysis period.

The average order value of approximately **₹27,621** indicates that each transaction generated a relatively high revenue value, largely influenced by higher-priced products.

4. Revenue Performance

4.1 Monthly Trend

January was the strongest month, generating approximately **₹5.21 million** in revenue.

Revenue subsequently fluctuated throughout the analysis period, with a particularly important decline occurring between May and June.

Key Finding

May → June revenue declined by approximately 22%.

June revenue was approximately **₹3.81 million**.

This decline should be treated as an important management signal rather than simply a monthly fluctuation.

Business Question

Management should investigate:

- Which products experienced the largest decline?
 - Which categories contributed to the decline?
 - Did specific cities experience a significant drop?
 - Were high-value products sold less frequently?
 - Was the decline caused by demand, pricing or product availability?
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5. Category Performance

The analysis shows significant concentration in category-level revenue.

Revenue Contribution

Electronics: approximately **78.2%**

Fashion: approximately **4.2%**

Books: approximately **0.3%**

Electronics is therefore the primary revenue driver.

Business Implication

The strong Electronics contribution represents an opportunity because the category is highly productive.

However, it also creates **category concentration risk**.

If Electronics sales decline significantly, total company revenue could be affected substantially.

Recommended Action

Management should:

- Protect availability of high-performing Electronics products.
 - Monitor Electronics sales trends closely.
 - Develop additional categories to reduce revenue dependency.
 - Identify products in lower-performing categories with growth potential.
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6. Product Performance

Revenue is also highly concentrated at the product level.

The top five products contributed approximately:

88% of total revenue.

The major revenue-generating products included:

- Laptop
- Tablet
- Smartphone
- Air Fryer
- Watch

Laptop was the highest-revenue product, generating approximately **₹11.41 million**.

Business Implication

The strong contribution from a small number of products means that these products are strategically important.

However, excessive dependence on a few products creates risk if:

- Inventory becomes unavailable.
- Product demand declines.
- Competitors introduce alternatives.
- Pricing pressure reduces margins.

Recommended Action

Maintain strong availability and monitoring of top products while increasing promotion of mid-tier products.

7. Geographic Performance

City-level analysis revealed meaningful differences in sales performance.

Leading Cities

The strongest-performing cities included:

- Pune
- Mumbai
- Jaipur

- Bangalore

Each generated approximately **₹3.1–₹3.3 million** in revenue.

Lower-Performing Cities

Hyderabad and Ahmedabad were approximately **35–40% below the leading city revenue levels**.

Business Implication

The performance gap suggests that there may be untapped growth opportunities in weaker markets.

Possible reasons include:

- Lower demand
- Product assortment differences
- Lower marketing activity
- Distribution limitations
- Pricing differences

Recommended Action

Conduct a city-level diagnostic for underperforming markets before increasing marketing or operational spending.

8. Revenue Concentration Risk

One of the most important findings is the concentration of revenue.

Approximately:

78.2% of revenue → Electronics

and approximately:

88% of revenue → Top five products

This creates a significant dependency on a limited number of revenue sources.

Business Risk

A decline in one major product or category could have a disproportionate impact on total revenue.

Recommended Strategy

The business should gradually diversify revenue by:

- Promoting mid-performing products.
 - Expanding successful products into weaker cities.
 - Improving low-contribution categories.
 - Testing targeted campaigns for Fashion and Books.
 - Monitoring product-level revenue concentration regularly.
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9. Pricing and Sales Volume Analysis

The analysis identified the following relationships:

Relationship	Correlation
Price vs Total Revenue	0.88
Quantity vs Total Revenue	0.27
Quantity vs Price	0.024

The strong relationship between price and transaction revenue is expected because transaction revenue is directly influenced by product price.

Quantity has a weaker relationship with transaction revenue.

Business Interpretation

Revenue growth should not be evaluated only through the number of units sold.

A smaller number of high-value products can generate substantially more revenue than a larger number of low-priced products.

Therefore, management should monitor:

- Revenue
- Units
- Orders
- Average Order Value
- Product contribution

together.

10. Short-Term Revenue Forecast

A Random Forest forecasting model was developed using historical revenue patterns and calendar features.

The model achieved:

Metric	Result
MAE	₹92,069
RMSE	₹108,779
MAPE	188.17%

The model was selected because it produced the lowest error among the evaluated approaches.

Seven-Day Forecast

Date	Expected Revenue
30 Jun 2026	₹312,523.78
1 Jul 2026	₹266,568.16
2 Jul 2026	₹242,149.56
3 Jul 2026	₹142,850.30
4 Jul 2026	₹106,184.20
5 Jul 2026	₹97,034.30
6 Jul 2026	₹100,033.94

Forecast Summary

Expected 7-day revenue: ₹1.27M

Average daily forecast: ₹181K

Highest forecast: ₹312.5K

Lowest forecast: ₹97.0K

The forecast should be used as a **planning indicator**, not a guaranteed revenue target.

11. Key Business Risks

The analysis identifies four major business risks.

Risk 1 — Product Concentration

The top five products generate approximately 88% of revenue.

Risk: Poor performance of one major product could significantly affect total revenue.

Risk 2 — Category Concentration

Electronics generates approximately 78.2% of revenue.

Risk: Heavy dependence on a single category increases exposure to category-specific demand changes.

Risk 3 — Geographic Performance Gap

Some cities perform substantially below the leading markets.

Risk: Potential revenue opportunities may remain underdeveloped.

Risk 4 — Revenue Decline

Revenue declined approximately 22% from May to June.

Risk: Continued decline could negatively affect future revenue performance.

12. Business Opportunities

Opportunity 1 — Expand Mid-Tier Products

The business can reduce product concentration by promoting products outside the current top five.

Opportunity 2 — Improve Underperforming Cities

Cities with lower revenue can be targeted for further analysis and controlled growth initiatives.

Opportunity 3 — Diversify Categories

Fashion, Books and other lower-contribution categories can be evaluated for targeted growth.

Opportunity 4 — Use Forecasting for Planning

Short-term forecasts can support:

- Inventory planning
 - Sales monitoring
 - Campaign planning
 - Revenue target setting
 - Management reporting
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13. Recommended Action Plan

Priority	Action	Business Purpose
High	Investigate May–June decline	Identify revenue slowdown drivers
High	Monitor top Electronics products	Protect major revenue sources
High	Analyze underperforming cities	Identify geographic growth opportunities
Medium	Promote mid-tier products	Reduce product concentration
Medium	Evaluate Fashion and Books	Increase category diversification
Medium	Monitor AOV and product contribution	Improve revenue quality
Medium	Use seven-day forecast	Support short-term planning

14. Management Dashboard KPIs

Management should monitor the following KPIs regularly:

Revenue KPIs

- Total Revenue
- Monthly Revenue
- Month-over-Month Growth
- Revenue by Category
- Revenue by Product

Sales KPIs

- Total Orders
- Total Units
- Average Order Value
- Average Selling Price

Geographic KPIs

- Revenue by City
- Revenue by Region
- City Growth Rate

Product KPIs

- Top Products
- Product Revenue Contribution
- Product Concentration

Forecasting KPIs

- Forecast Revenue
 - Actual vs Forecast Revenue
 - Forecast Error
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15. Final Business Insights

The analysis leads to five major conclusions:

1. Electronics is the primary revenue engine.

With approximately 78.2% of revenue, Electronics has the greatest influence on overall business performance.

2. Revenue is highly concentrated.

The top five products contribute approximately 88% of total revenue.

3. Geographic performance is uneven.

Leading cities generate significantly more revenue than some other markets, creating potential expansion opportunities.

4. June requires management attention.

The approximately 22% decline from May to June should be investigated to determine whether it represents a temporary fluctuation or an emerging trend.

5. Forecasting can support short-term planning.

The seven-day forecast provides an additional analytical signal that management can use alongside actual sales performance.

16. Conclusion

The e-commerce sales analysis provides a clear view of business performance across products, categories, cities and time.

The most important finding is that the business currently depends heavily on **Electronics and a small group of high-performing products**. This concentration supports strong revenue generation but also introduces business risk.

The decline in June revenue and the performance gap between cities provide opportunities for further investigation.

Management should prioritize protecting high-performing products while gradually diversifying revenue across products, categories and geographic markets.

The forecasting model can complement traditional reporting by providing a short-term view of expected revenue. However, forecasts should be combined with actual sales data and business context before making operational decisions.

Overall recommendation:

Protect the current revenue drivers, investigate the June slowdown, improve underperforming markets, diversify product/category contribution, and use forecasting as an additional planning tool.